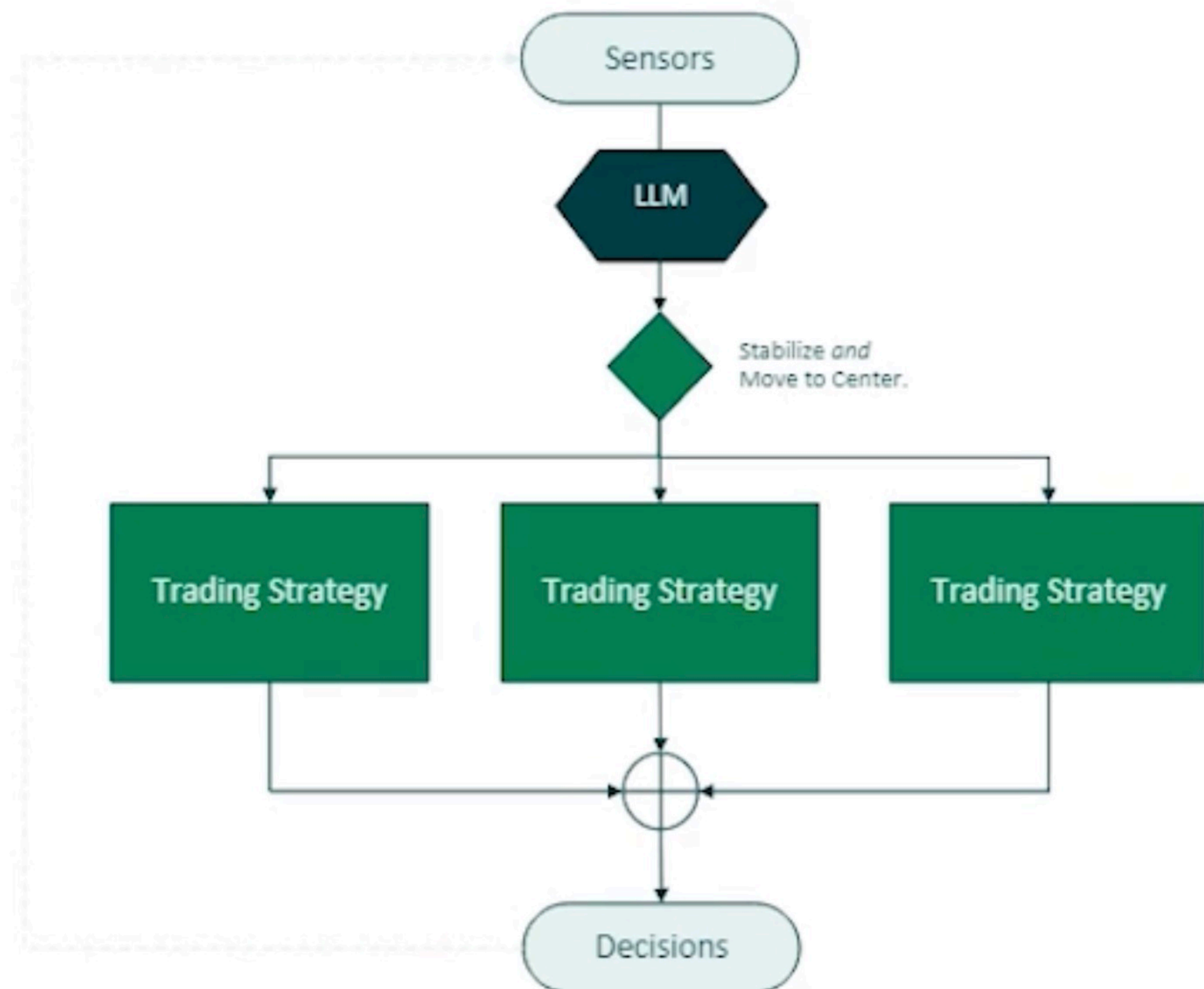
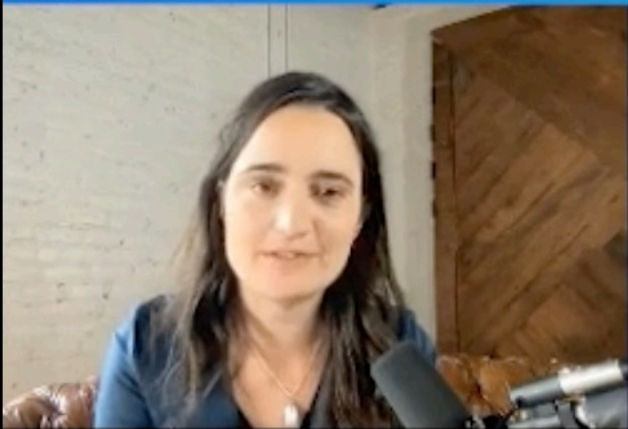


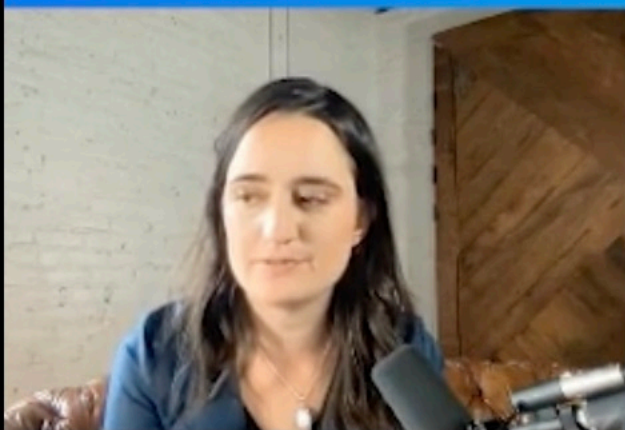
Design Patterns for intelligent autonomous agents: Research Pattern



A small video inset in the top-left corner shows Jennifer Kaehms, a woman with long dark hair, wearing a blue top, speaking into a microphone.

Intro

- My name is Jennifer Kaehms (@jennykaehms)
- Investing since 2016 – Luminar, Lambda Labs
- Worked at Forethought AI
- Now a partner at Foundation Capital



We've invested in multiple eras of AI starting in 2010

RULES
BASED

MACHINE
LEARNING
ERA

DEEP
LEARNING
ERA

LARGE
LANGUAGE
MODEL ERA

1960

1970

1980

1990

2000

2001

2006

2009

2012

2013

2014

2015

2016

2017

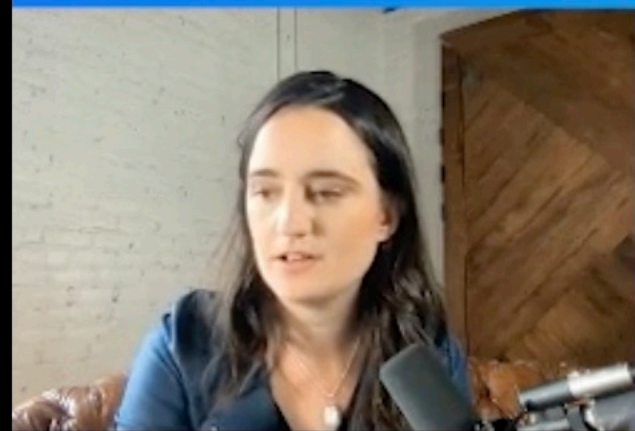
2018

2019

2020

2021

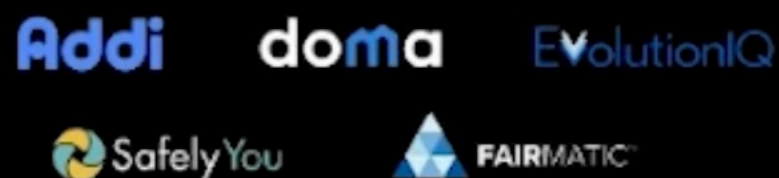
2022



FC AI Portfolio

Intelligent Automation

AI VERTICAL APPS



AI HORIZONTAL APPS



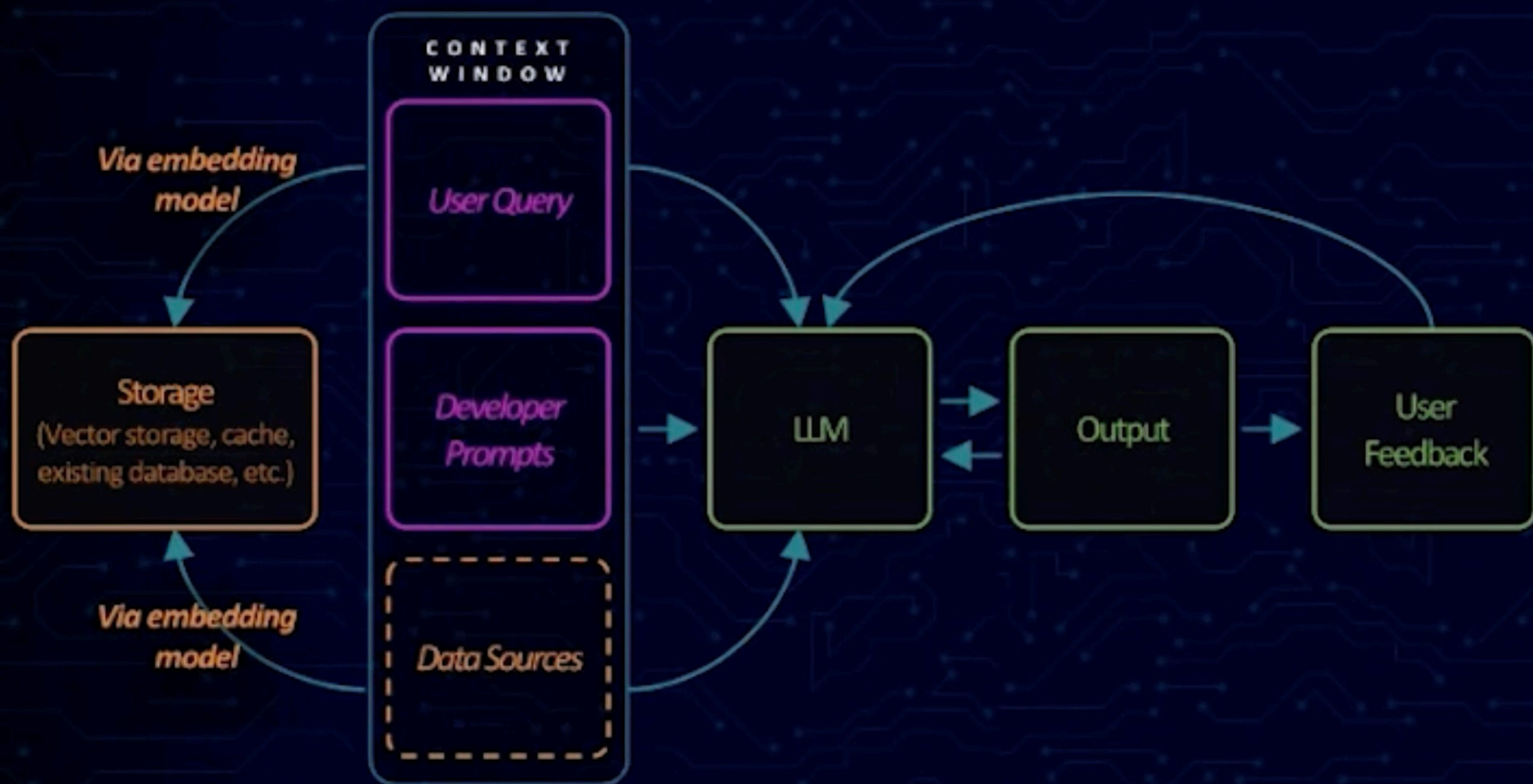
Data / ML Infrastructure



Data Security



Generative AI model: automates one task in your application



1. **Prompt Engineering:**
Ask better questions

2. **Retrieval:**
Add more context for better responses

3. **Finetuning**
Train model to give better responses through continuous iteration

→ Outerbounds

→  anyscale

LLM Observability

 arize

Security

 Fortanix

 skyflow

One context window

AI Agents

Browser Buddies

Returns an answer, not a list of links (ChatGPT)

Helps you browse the internet & take actions (OpenAI Plugins & MultiOn)

Social Media

Creates a photo for your social media (Midjourney or Lensa)

Generates a game for you to play & interact with

Legal Work

Searches for case files (Casetext)

Processes case information & generate filings

One context window

AI Agents

Healthcare

Returns information based on query

Helps you maintain your insulin level

Education

Suggests answers to test questions & homework

Generates a lesson plan that adapts to your learning goal

Logistics

Fills out a customs form

Plans route & navigates to your desired destination

Types of Agents

Reflex
based

World-
model
based

Goal
based

Utility
based

Object
centric or
"curious"

Agent Adaptability



One context window

AI Agents

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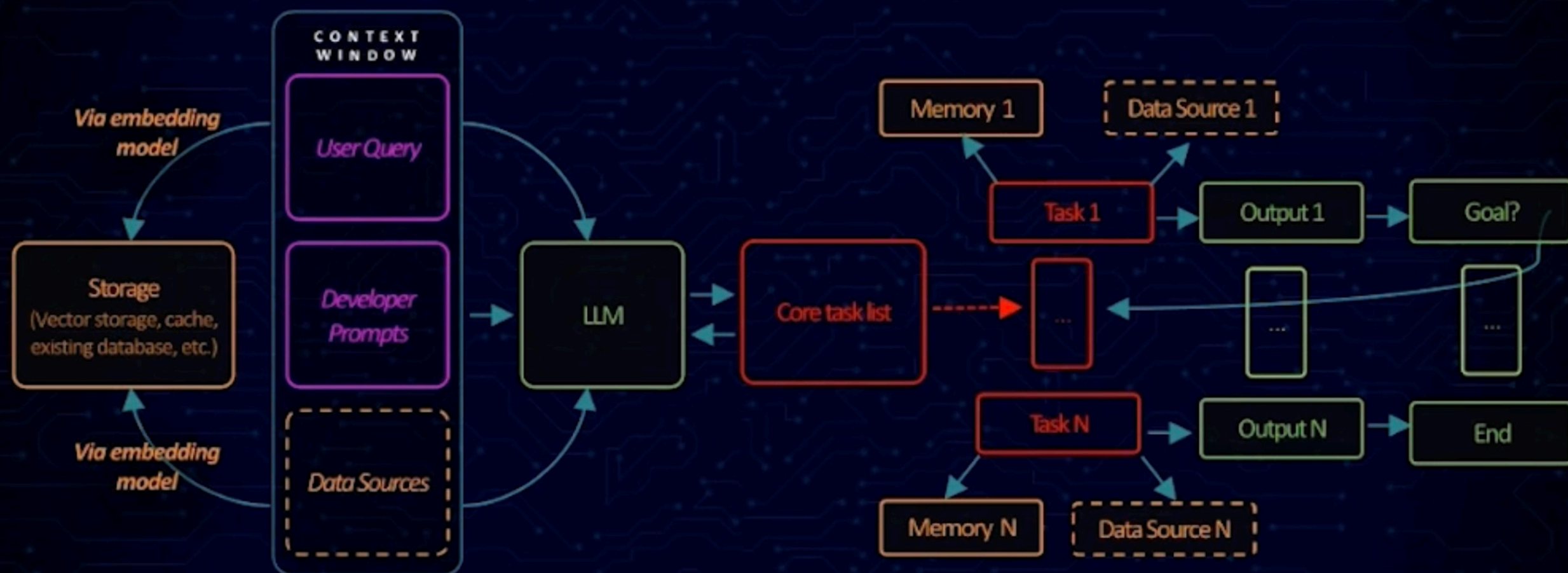
Fills out a customs form

Plans route & navigates to your desired destination

Utility based

Building AI Agents with LLMs

Generative AI agent: automates a sequence of tasks in your application



LLM Observability

arize

Security

Fortanix skyflow

Building Your First AI Agent

1.

Why agents?

Why would you need an agent?

2.

**Two questions to ask
before starting an
agent project**

- 1) Is this system isolated to one app?
- 2) Is this a chatbot?

3.

Demo

Automating my job with an AI agent

AI Agents for Internal Use Cases

Is the workflow isolated to one system/app?

NO

Is it deterministic and automatable using set rules and APIs to move data between the different systems?

NO

Consider if the bottlenecks for automating this system could be solved by an AI agent.

YES

Traditional workflow building tools like Retool, Zapier will likely work here.

YES

Does automating use of this system require expertise the provider won't support with their own AI features?

NO

Tell them to build the functionality you need :)

YES

Consider setting up an RPA to connect this system to a well-trained LLM. (Layer)

AI Agents for External Use Cases

Is this a chatbot?

NO

Is the LLM-based feature you want to build very specific to your domain, business or product?

NO

Checkout new SaaS products created to do LLM-based tasks.

YES

If it's summarization, doc synthesis, or similarity search, you can likely build it yourself! Build an AI agent.

YES

With a decent prompt and example data in ChatGPT, can you get ChatGPT to work pretty well?

NO

If your chatbot truly requires domain expertise or input output pairs outside of OpenAI training data, consider training your own model.

YES

Consider security risks, but otherwise, any decent SWE should be able to build this in a product cycle or 2. Or embed a SaaS tool.

Final Thoughts

Method

Challenge & Restrictions Today

Considerations

Choosing an Agent Architecture

- Reflex based
- World-model based
- Goal based
- Utility based
- Object centric or "curious"

- The more autonomy an agent has, the more likely it is to drift
- Creating model-based agents requires an accurate model of the world
- As you connect agents to more data sources, you must consider data security & privacy

- When creating a solution for your org, choose the simplest solution possible
- It's possible to have multiple agents work together to learn & achieve goals

Security

- Embeddings models (open source or proprietary)
- Vector search
- Retrieval methods

- Privacy concerns: storing customer schema / data information with vector database third parties indefinitely
- Retrieval results are typically poor; high dimensionality, sparse representation, domain adaptation, out-of-vocabulary words, language coverage
- Latency, hallucination, reliability challenges

- More of a "data engineering" problem vs an ML problem
- Need to understand what data is relevant & how to chunk data appropriately

Reliability and Accuracy

- Chain of thought
- Curious replay
- Object centric

- Can fail to generate results when inputs are vague
- Model loops on a task
- The "Binding Problem"

- How well does the model do on the Criter Benchmark
- For workflow tasks, chain of thought works

Building Human-like AI Agents

Solving challenges around Computer Interaction, Long-Term Memory, and Agent-Agent Communication

Div Garg (MultiOn AI)

Introduction

- Actions and Emergent Agent Architectures
- Building Human-like AI Agents
- Computer Interactions using AI
- Long-Term Memory and Personalization
- Agent to Agent communication
- Future Directions for Autonomous AI Agents



Building AI Agents

1. Why?
2. How?
3. Ingredients?
4. What can they do?



Key thesis: Humans will communicate with AI using natural language and AI will operate machines allowing for more intuitive and efficient operations

Software 3.0

AI Agents

1. Why?

A single call to a large Foundation AI model is not enough. A lot more can be unlocked by building *AI systems*

1. How?

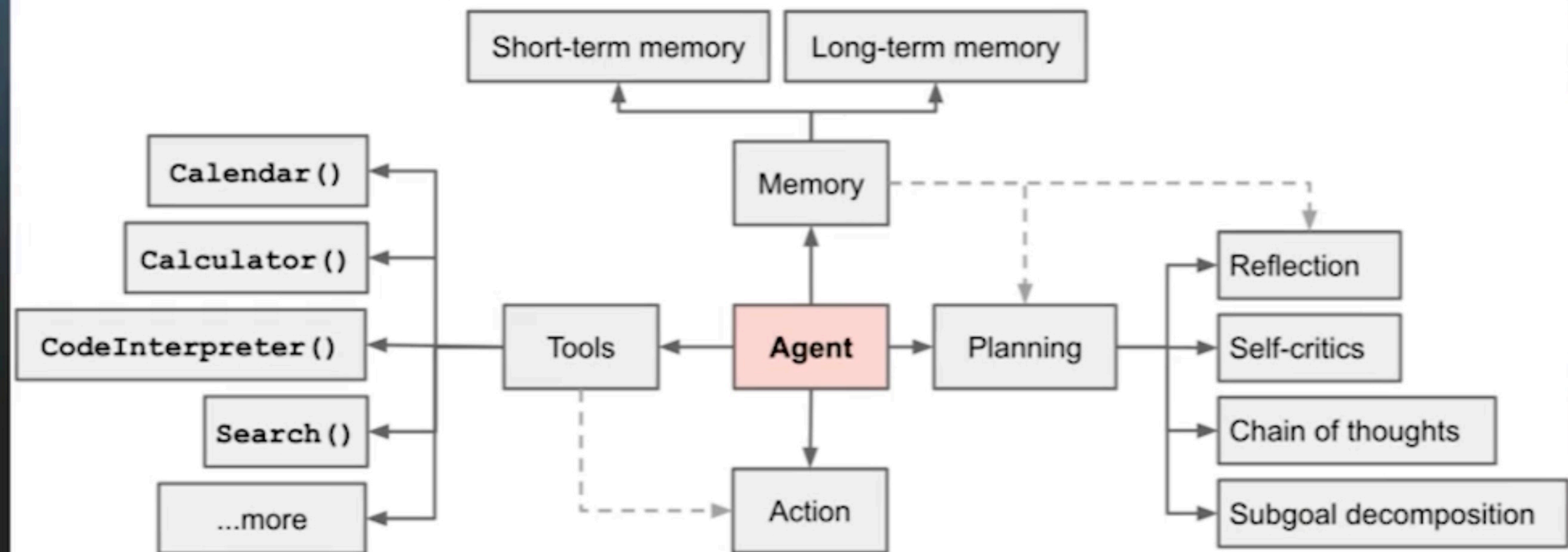
Using model chaining, reflection & other mechanisms

1. Ingredients?

Memory, context length, personalization, actions, internet access...

1. What can they do?

AI Agents





MULTI-ON

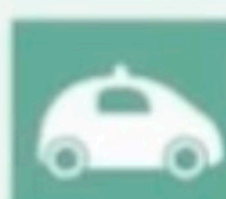
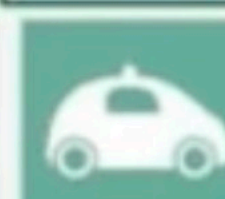
- **Multiple Utility AI Language Tool for Internet Operations and Navigation (MULTION)**
- **Agent with full R/W access to the internet**

Name Origin: from Quantum Physics (neutron, muon, fermion, ...)

Why human-like AI Agents?

- 1. **Can do what you can do:** Able to use existing interfaces designed for humans & operate outside programmatic boundaries
- 1. **Digital extension of you:** Can act as an extension of the user and act on their behalf
- 1. **Less-restrictive boundaries:** Can handle logins, payments, etc. and interact with services without any API restrictions
- 1. **Simple action space:** Need only click & type action primitives
- 1. **Self-learning:** Can learn from the user and self-improve with more interactions

5 levels of Autonomy

For on-road vehicles			 Human driver	 Automated system		
			Steering and acceleration/ deceleration	Monitoring of driving environment	Fallback when automation fails	Automated system is in control
Human driver monitors the road	0	NO AUTOMATION				N/A
	1	DRIVER ASSISTANCE				SOME DRIVING MODES
	2	PARTIAL AUTOMATION				SOME DRIVING MODES
Automated driving system monitors the road	3	CONDITIONAL AUTOMATION				SOME DRIVING MODES
	4	HIGH AUTOMATION				SOME DRIVING MODES
	5	FULL AUTOMATION				

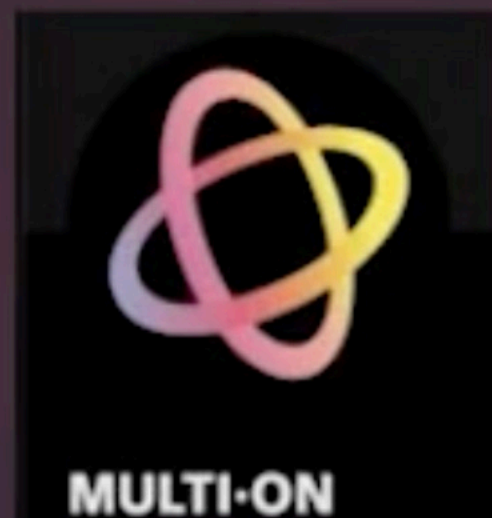
Agent Computer Interaction

Two routes



**API
(programmatic)**

**Direct interaction
(browser or desktop control)**



easy to build context
safer & controllable
high variability

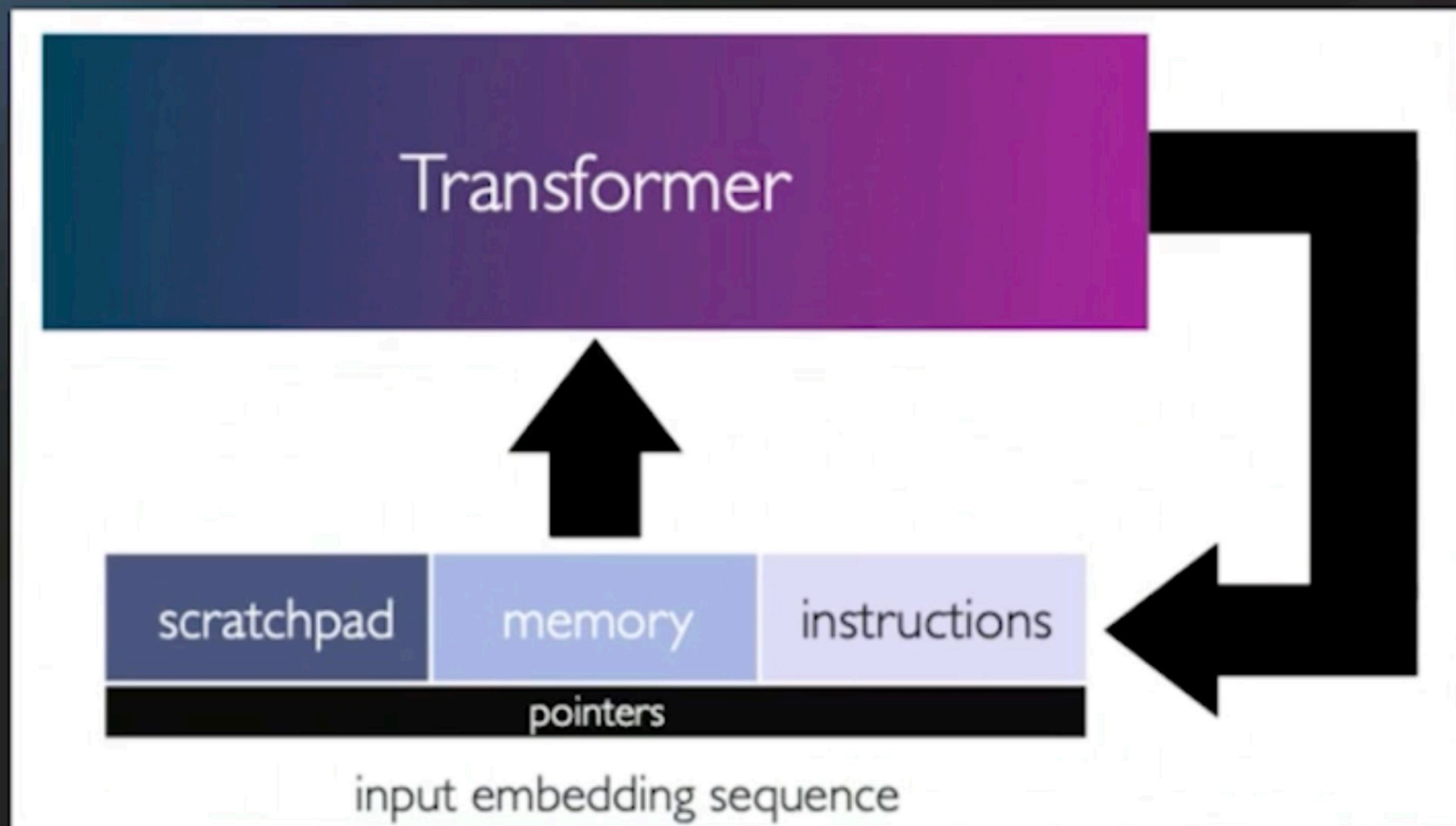
easy to take actions
free-form interactions
need to provide guarantees



ChatGPT Plugins



AI Models as Neural Compute Unit



Looped Transformers

Long-term Memory

- **Works similar to disk (long-lived & persistent)**
- **Mechanisms**
 - Embeddings
 - Retrieval models
- **Open Questions:**
 - Hierarchy
 - Temporal Coherence
 - Structure
 - Online adaptation



Personalization

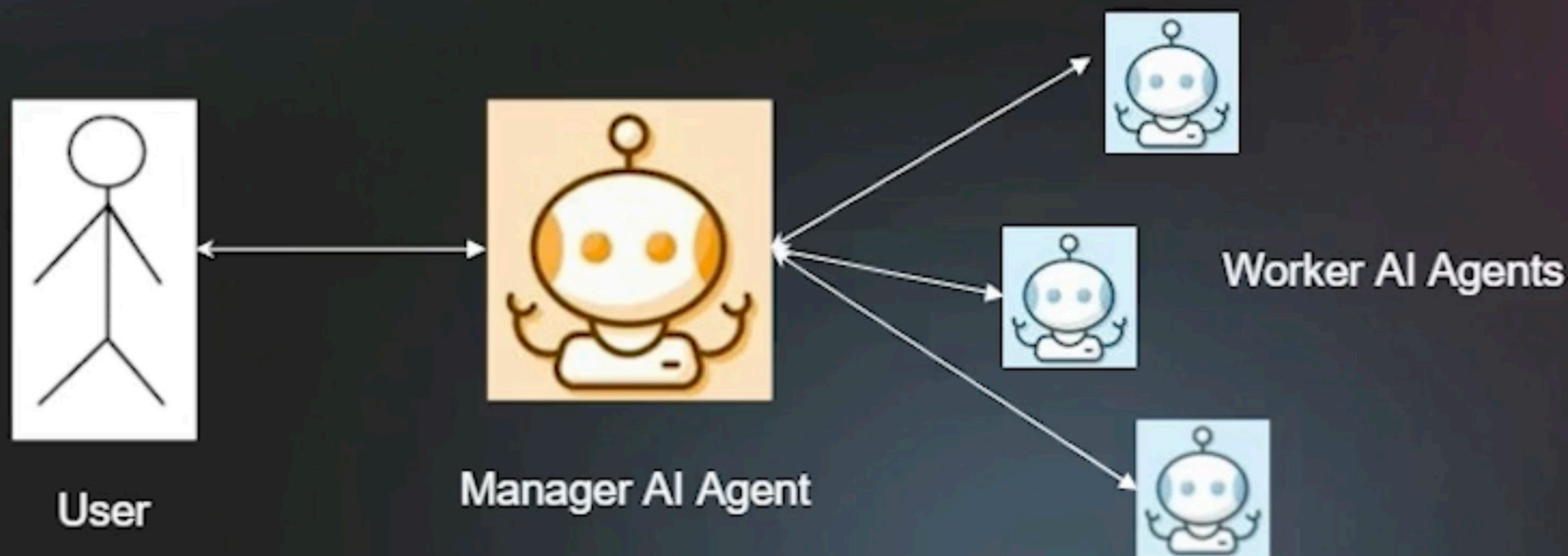
- **Agent-user Alignment:** Enable agent to take actions that are aligned with the user preferences
- Everyone has different prefs & likes/dislikes:
 - **Explicit:** allergies, favourite dishes, flight seat prefs, ...
 - **Implicit:** choice between brands, out of 10 items in a listing which user likes better

Challenges

- **Collecting user data & preferences:**
 - actively asking for preferences
 - passive learning from interactions
- **Learning from user preferences: supervised fine-tuning vs human-feedback**
- **On-fly adaptation**
- **Privacy**

Agent to Agent Communication

- **Exchanging info between fleets of agents**
- **Hierarchies**
- **Syncing primitives**



Agent to Agent Communication

- Robust communication protocols & Syncing primitives

Scenario 2:
Task was incorrectly done
(Agent Miscommunication)

Verify Task X:
- required spec

Worker State

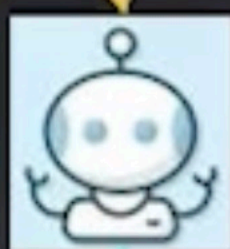
Task X:
(status: **done**)

Manager state

Task X:
(status: **verify done**)



Manager AI Agent



Verify Response:
- actions were not correct ✗

Agent to Agent Communication

- Robust communication protocols & Syncing primitives

Scenario 2:
Task was incorrectly done
(Agent Miscommunication)

Re-do Task X:

- plan
- context
- feedback/corrections

Worker State

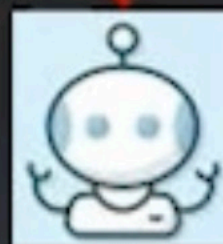
Task X:
(status: **not done**)

Manager state

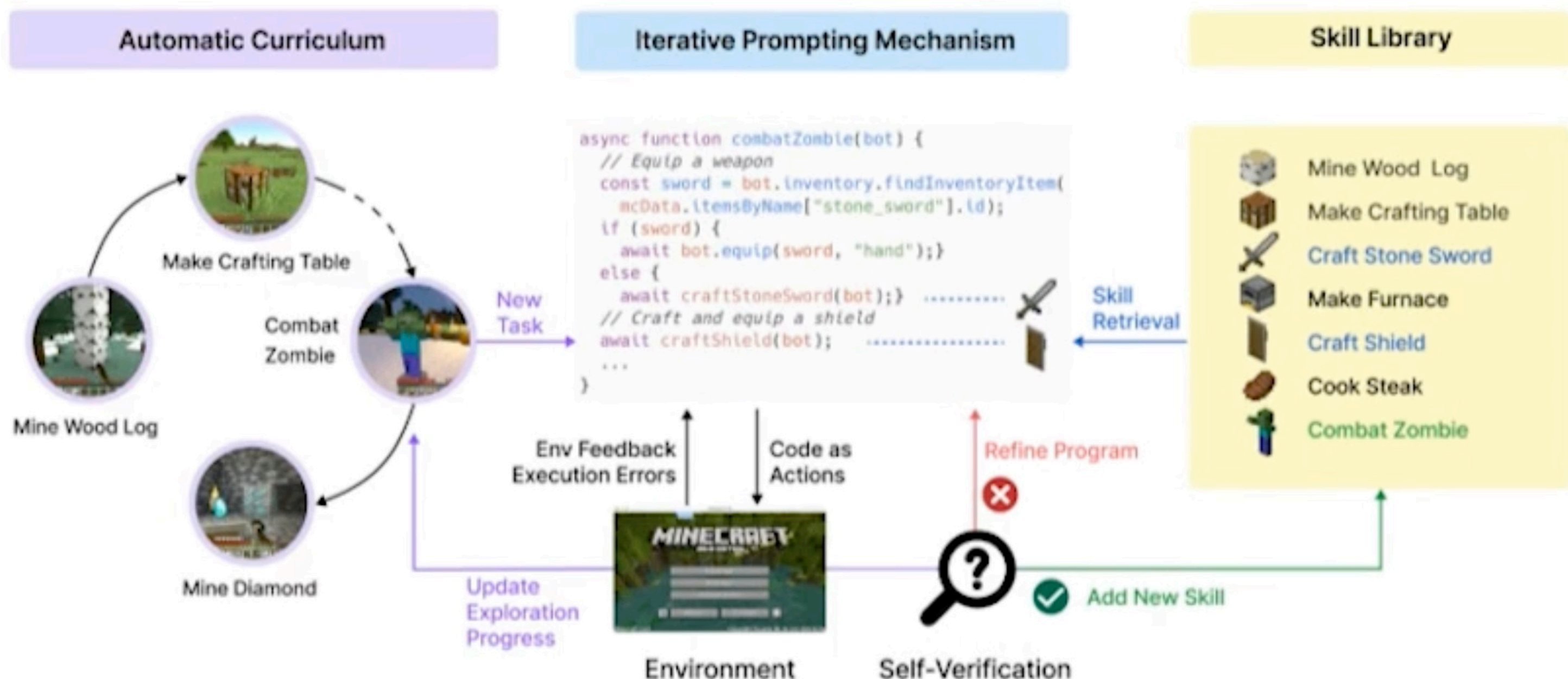
Task X:
(status: **not done**)



Manager AI Agent



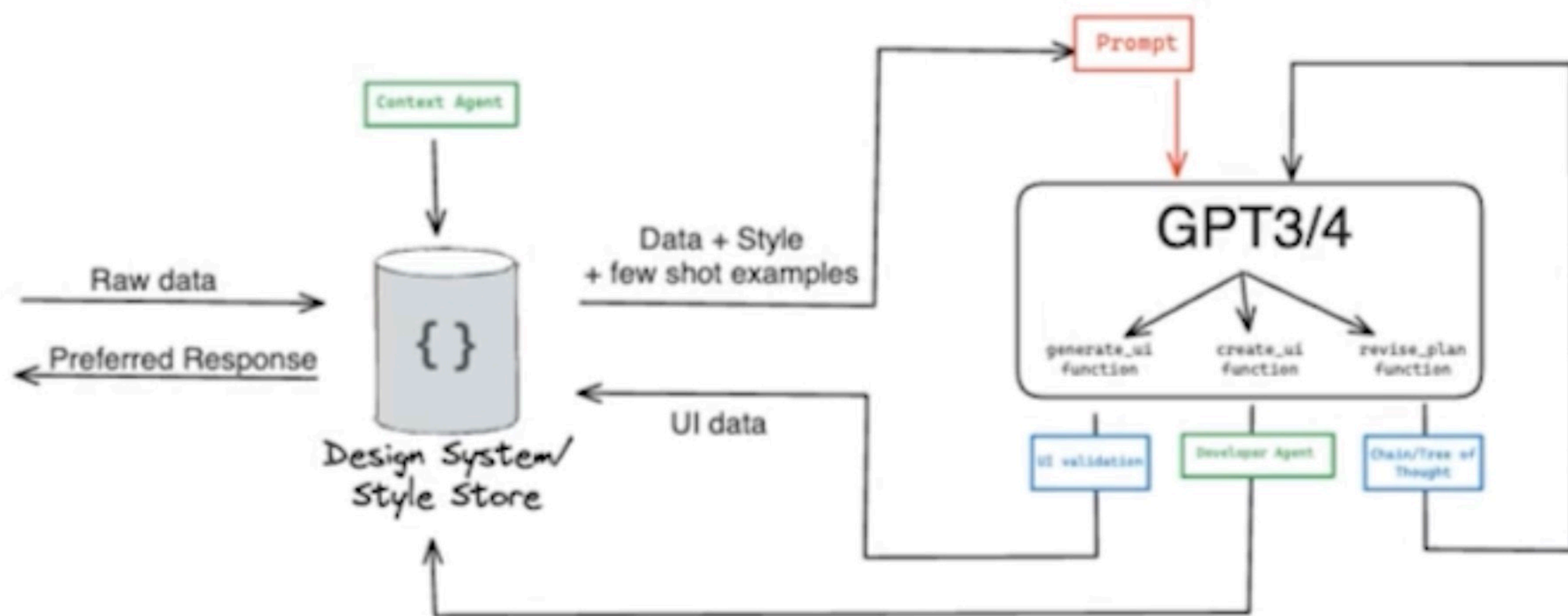
Voyager



Voyager consists of three key components: an automatic curriculum for open-ended exploration, a **skill** library for increasingly complex behaviors, and an iterative prompting mechanism that uses code as action space.

Building a Interface Agent

Interface Agent



Building AI Agents with LLMs (Large Language Models)

Patterns

- Copilot Pattern
- Research Pattern

- AutoGPT

Selector

- Rule Based Selector
- Learning Based Selector

Skills (Decision Maker)

OpenAI

- Get Info
- Tech Crunch
Hacker News

References

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- <https://github.com/kenceanderson>
- <https://github.com/MrKaehms>

Div Garg (Multi0n AI)

https://en.wikipedia.org/wiki/Latent_space

Arjun Bansal